

A project report
on
**Multi Behaviour Analysis of Dairy Cows for
Thermotolerance Assessment**

submitted in partial fulfilment of the requirements for the degree of

MSc Computer Science

in

Department of Computer Science and Applications

by

Rahul Bidyasagar
(Roll No: 03S24CS02)

under the guidance of

Prof. Chandra Sekhar Panda



Department of Computer Science and Applications
Sambalpur University, Jyoti Vihar, Burla,
Sambalpur, Odisha- 768019

2024-26



Department of Computer Science and Applications
Sambalpur University, Jyoti Vihar, Burla,
Sambalpur, Odisha- 768019
Tel: (0663)2431016, Website: www.suniv.ac.in

Date: __/04/2026

Undertaking

I declare that the project work presented in this report titled “**Multi behaviour Analysis of Dairy cows For Thermotolerance Assessment**”, submitted to the Department of **Computer Science and Applications, Sambalpur University, Odisha**, for the award of the **M. Sc. in Computer Science**, is my original work. I have not plagiarised or submitted the same work for the award of any other degree. In case this undertaking is found incorrect, I accept that my degree may be unconditionally withdrawn.

Rahul Bidyasagar
Roll: 03S24CS02



Department of Computer Science and Applications
Sambalpur University, Jyoti Vihar, Burla,
Sambalpur, Odisha- 768019
Tel: (0663)2431016, Website: www.suniv.ac.in

Date: 30/04/2026

Supervisor's Certificate

This is to certify that the project work in the report entitled **Multi behaviour Analysis of Dairy cows For Thermotolerance Assessment** submitted by **Rahul Bidyasagar** is a record of an original research work carried out by him under my supervision and guidance in partial fulfilment of the requirements for the award of the degree of MSc Computer Science. Neither this report nor any part of it has been submitted for any degree or academic award elsewhere.

Prof. Chandra Sekhar Panda
Professor & Hod

Department of Computer Science & Application
Sambalpur university



Department of Computer Science and Applications
Sambalpur University, Jyoti Vihar, Burla,
Sambalpur, Odisha- 768019
Tel: (0663)2431016, Website: www.suniv.ac.in

Date: 30/04/2026

Certificate of Examination

Roll:03S24CS02

Name: Rahul Bidyasagar

Title of Project: Multi behaviour Analysis of Dairy cows For Thermotolerance Assessment

We the below signed, after checking the report mentioned above and the official record book(s) of the student, hereby state our approval of the dissertation submitted in partial fulfilment of the requirements of the degree of **Master of Science in Computer Science** in the department of **Computer Science and Applications, Sambalpur University**, Odisha. We are satisfied with the volume, quality, correctness, and originality of the work done by the student.

Prof. Chandra Sekhar Panda
Head, Department of Computer
Science and Applications
Sambalpur University

External Examiner

Internal Examiner

Acknowledgements

I would like to express my deepest and most sincere gratitude to my Guide, **Prof. Chandra Sekhar Panda Sir, Head of the Department**, for his invaluable guidance, constant encouragement, and unwavering support throughout the completion of this project.

I am also sincerely thankful to the **Department of Computer Science and Applications, Sambalpur University**, for providing a supportive academic environment and the necessary resources to successfully carry out this project.

I extend my heartfelt thanks to all the faculty members of the department for their guidance and knowledge, which have contributed significantly to my academic growth.

I would like to acknowledge the creators of the MMCOWS dataset for making such a comprehensive multimodal dataset available for research, which greatly supported the implementation of this work.

Finally, I would like to thank my friends and classmates for their support, discussions, and encouragement, which made this journey more meaningful and enjoyable.

Rahul Bidyasagar

Abstract

Heat stress is a significant challenge for global dairy cattle production, both economically and physiologically. As temperatures and humidity levels rise due to climate change, the ability to quickly and accurately assess thermotolerance becomes essential for sustainable livestock management. Thermotolerance refers to an animal's ability to maintain productivity and physiological stability during heat stress. This research offers a machine learning framework for evaluating multiple behaviours in dairy cows to assess thermotolerance, utilizing the MMCOWS multimodal dataset (NeurIPS 2024).

The study combines core body temperature (CBT) data from vaginal temperature sensors, behavioural observations from camera-based computer vision, and environmental factors such as the Temperature-Humidity Index (THI) and relative humidity, all from MMCOWS. A feature engineering pipeline creates composite thermal-behavioural features, while class imbalance is addressed using the Synthetic Minority Over-sampling Technique (SMOTE). The study tests thirteen classifiers, including instance-based learning, decision trees, ensemble methods, and deep learning architectures, to predict heat stress levels: Normal ($\text{THI} < 72$), Mild ($72 \leq \text{THI} < 78$), and Severe ($\text{THI} \geq 78$). Among the models tested, K-Nearest Neighbours (KNN) achieves the highest accuracy at 79.43% (macro F1: 79.35%), followed by Decision Tree at 77.17% and XGBoost at 74.73%. One-way ANOVA confirms significant differences in core body temperature across the labelled behaviour groups ($p < 0.05$), supporting the idea that behavioural signals can serve as meaningful predictors of thermal stress.

The proposed pipeline delivers strong predictive accuracy and shows that combining physiological sensor data with behaviour labels significantly outperforms single-modality approaches. This research aids the field of Precision Livestock Farming (PLF) by offering a reproducible, data-driven method for non-invasive, automated thermotolerance monitoring in commercial dairy settings.

Keywords: Dairy cattle, heat stress, thermotolerance, behaviour analysis, XGBoost, SMOTE, MMCOWS, Temperature-Humidity Index, precision livestock farming, machine learning.

Contents

- 1. Introduction 1
 - 1.1 Background and Motivation..... 1
 - 1.2 Problem Statement 1
 - 1.3 Objectives..... 2
 - 1.4 Scope and Limitations 2
- 2. Literature Review 3
 - 2.1 Precision Livestock Farming and Machine Learning..... 3
 - 2.2 Behavioural Indicators of Heat Stress 3
 - 2.3 Multimodal Sensing for Cattle Monitoring 4
 - 2.4 The MMCOWS Dataset 4
- 3. Dataset Description..... 6
 - 3.1 Overview of MMCOWS 6
 - 3.2 Sensing Modalities Used in This Study 6
 - 3.3 Data Structure and Annotation 7
 - 3.4 Visual Representation of the MMCOWS Dataset..... 7
- 4. Methodology..... 9
 - 4.1 System Architecture 9
 - 4.2 Data Loading and Preprocessing..... 9
 - 4.3 Timestamp Alignment and Merging 9
 - 4.4 Feature Engineering 10
 - 4.5 Target Variable: Stress Level Classification 11
 - 4.6 Class Imbalance Handling with SMOTE 11
 - 4.7 Model Training and Evaluation..... 11
- 5. Experimental Results..... 13
 - 5.1 Descriptive Statistics by Behaviour and Stress Level 13
 - 5.2 Model Performance Comparison..... 14
 - 5.3 Statistical Analysis: ANOVA on Behaviour vs. Body Temperature 16
 - 5.4 Visualization of Results 17
- 6. Discussion..... 20
- 7. Conclusion and Future Work..... 23
- References 24

List Of figures

Figure 1 : MMCOWS Sensor Overview 6

Figure 2 : Behaviours Overview..... 7

Figure 3 : MMCOWS data processing workflow (Adapted from Vu et al., 2024) 8

Figure 4 : 3D visualization of the MMCOWS multimodal sensing and four camera views..... 8

Figure 5 : Feature Importance (LightGBM) 10

Figure 6 : Model Performance Heatmap..... 15

Figure 7 : Classifier Accuracy Comparison..... 16

Figure 8 : CBT Distribution by Behaviour 17

Figure 9 : CBT vs THI (Scatter Plot) 18

Figure 10 : KNN Confusion Matrix..... 18

Figure 11 : Correlation Heatmap of Features and Stress Level..... 19

List of tables

Table 1: Summary of sensing modalities used in this study 6

Table 2 : Behaviour definitions of cows observed in isometric views 7

Table 3 : stress level classes based on Temperature-Humidity Index (THI)..... 11

Table 4 : Summary of Observed Counts Across Behaviour and Stress Levels..... 13

Table 5: Comparative evaluation of machine learning models 14

1. Introduction

1.1 Background and Motivation

The global dairy industry faces increasing pressure to improve sustainability in environmental, social, and economic areas. Dairy cattle, especially Holstein-Friesian breeds that dominate commercial production, are very sensitive to heat stress. This stress happens when an animal's heat load is greater than its ability to cool down. The result is a rise in core body temperature (CBT) beyond the thermal neutral zone. This triggers several changes, including less feeding time, longer periods of standing, lower milk production, and changes in drinking patterns.

The Temperature-Humidity Index (THI) combines ambient temperature and relative humidity to assess heat stress risk in livestock environments. A THI below 72 is considered thermoneutral. Values between 72 and 78 indicate mild stress, while values above 78 show severe heat stress conditions. These thresholds directly relate to observable changes in cattle behaviour, making behavioural monitoring an effective, non-invasive way to gauge their thermal state.

Precision Livestock Farming (PLF) has sparked the use of sensor technology and machine learning for ongoing, automated animal monitoring. By using wearable sensors, environmental loggers, and computer vision systems, PLF platforms allow for detailed assessments of animal health and welfare on an individual level. This approach changes herd management from a reactive to a proactive, data-driven process.

1.2 Problem Statement

Despite the availability of individual sensing modalities for cattle monitoring, most existing datasets and studies examine behaviours or physiological indicators in isolation. A critical gap exists in the literature for multimodal, temporally synchronized datasets that combine behavioural labels, physiological measurements, and environmental indices in a unified analytical framework. Furthermore, automated thermotolerance assessment—the ability to predict an individual animal's stress state from observable behavioural and sensor signals—remains underexplored in the context of multi-class classification with real-world, imbalanced data.

This study addresses these gaps by developing a machine learning pipeline that fuses behavioural annotations, core body temperature readings, and environmental data from the MMCOWS dataset to classify dairy cows into ordinal heat stress categories. The central research question is: Can a multi-behavioural, sensor-fused machine learning model reliably

predict THI-based stress levels in dairy cows, and do behavioural signals carry statistically significant discriminative information about physiological thermal state?

1.3 Objectives

To develop a machine learning-based system that integrates behavioural, physiological, and environmental data to accurately assess and classify heat stress levels in dairy cows.

1.4 Scope and Limitations

This study focuses on one annotated day (July 25, 2023) of the MMCOWS dataset, corresponding to the single day with full behavioural ground truth for all 16 cows. The multi-cow pipeline extends analysis across all cows for which both behavioural and CBT data are available. The scope is limited to the specific sensor modalities available in MMCOWS (CBT, THI, humidity, behaviour labels) and does not include UWB location data or image-based features. The classification framework is a supervised, batch-learning approach and does not address real-time streaming inference.

2. Literature Review

2.1 Precision Livestock Farming and Machine Learning

Precision livestock farming (PLF) has become an important approach to tackle sustainability challenges in the dairy industry. With the help of modern sensing, computing, and communication technologies, PLF allows farmers to monitor individual animals in real time. It helps track their behaviour, health, and productivity, making it easier to detect problems early and manage farms more efficiently. In recent years, PLF has advanced further with the use of machine learning techniques. Many approaches now combine computer vision and wearable sensors—for example, systems that estimate a cow's body weight using images or detect respiratory diseases using inertial measurement units (IMUs).

García et al. (2020) reviewed machine learning applications in PLF and identified more than 100 studies using methods such as neural networks, support vector machines, and ensemble models for tasks like behaviour recognition and health monitoring. Similarly, Mahmud et al. (2021) focused on deep learning in precision cattle farming, highlighting the use of convolutional neural networks for identifying cows and recognizing their posture. Among different approaches, ensemble tree-based models like gradient boosting and random forests have shown strong performance, especially for sensor-based tabular data, as they handle noisy data well and can capture complex relationships between features.

2.2 Behavioural Indicators of Heat Stress

Behavioural and physiological responses in dairy cattle are commonly used in both research and real farm settings to detect health problems, especially heat stress. When cows experience heat stress, they tend to stand more, which helps them lose heat through air circulation. They also eat less frequently to reduce internal heat production, which can lead to a drop in milk yield. In addition, heat-stressed cows usually drink water more often, but in smaller amounts each time.

Cook et al. (2007) found that heat stress and lameness can significantly change the daily activity patterns of lactating cows, with standing time increasing during periods of high Temperature-Humidity Index (THI). Allen et al. (2015) showed that core body temperature (CBT), time of day, and environmental conditions are closely linked to changes in cow behaviour, indicating that increases in CBT are often accompanied by noticeable behavioural changes. Later studies (Tresoldi et al., 2019; Hut et al., 2022) also confirmed that behaviour measured using sensors is strongly related to THI levels, supporting the idea that behavioural data can be used as a reliable indicator of heat stress in dairy cattle.

2.3 Multimodal Sensing for Cattle Monitoring

Behavioural and physiological sensing of dairy cattle has progressed substantially from single-modality studies to increasingly integrated approaches. Accelerometers mounted on the neck or ankle provide useful information about a cow's body movements and postures, while UWB systems provide precise location data that can be used to infer feeding, drinking, and social interaction activities. Vision-based approaches using RGB cameras eliminate the need for wearable devices, simplifying deployment and improving animal comfort, though they are susceptible to changes in lighting conditions and physical obstructions.

Core body temperature, measured via vaginal or rectal sensors, is considered the gold standard physiological indicator of heat stress due to its direct correspondence with thermal load. More recently, subcutaneous and ear-tag-based temperature sensors have been validated as fewer invasive alternatives. The combination of behavioural and physiological sensor streams has been shown to outperform either modality alone: Borghart et al. (2021) achieved improved lameness prediction by combining activity data with milk production records, illustrating the complementary value of sensor fusion in cattle health applications.

2.4 The MMCOWS Dataset

MMCOWS (Vu et al., NeurIPS 2024) is one of the most comprehensive publicly available multimodal datasets for dairy cattle monitoring. It includes data collected from sixteen milking Holstein cows over a period of two weeks at the University of Wisconsin–Madison’s Arlington Agricultural Research Station. Unlike earlier datasets that rely on only one or two types of sensors, MMCOWS combines multiple data sources, including UWB-based location tracking, inertial and magnetic sensors, air pressure measurements, vaginal temperature sensors, ankle sensors, multi-view RGB cameras, indoor microclimate sensors, outdoor weather data, and daily milk yield records.

The dataset contains around 4.8 million image frames, along with one fully annotated day that includes approximately 213,000 bounding boxes for all 16 cows. These annotations cover seven behaviour classes: walking, standing, feeding head up, feeding head down, licking, drinking, and lying. Benchmark studies using this dataset show that combining multiple sensor modalities especially UWB with head direction and ankle data significantly improves performance compared to single-modality approaches, achieving F1 scores above 0.73 for behaviour classification.

A key strength of the MMCOWS dataset is its **real-world data collection setup**, where sensors were deployed in a working dairy barn without disturbing the natural

behaviour of the animals. This ensures that the data reflects realistic farm conditions rather than controlled laboratory environments. Additionally, all sensing modalities are **time-synchronized**, allowing accurate alignment between behavioural, physiological, and environmental data streams. This temporal consistency is crucial for studying cause–effect relationships, such as how changes in temperature and humidity influence behaviour and core body temperature over time.

Another important aspect of MMCOWS is its **multimodal fusion capability**, which enables researchers to explore relationships across different data types. For example, location data from UWB can be combined with behavioural annotations and temperature readings to better understand movement patterns under heat stress conditions. The dataset also includes both **primary data (raw sensor readings)** and **secondary processed data**, such as head direction and posture classification, making it suitable for a wide range of machine learning applications.

Overall, the richness, scale, and multimodal nature of MMCOWS make it highly suitable for developing advanced machine learning models for thermotolerance assessment, behaviour analysis, and precision livestock farming.

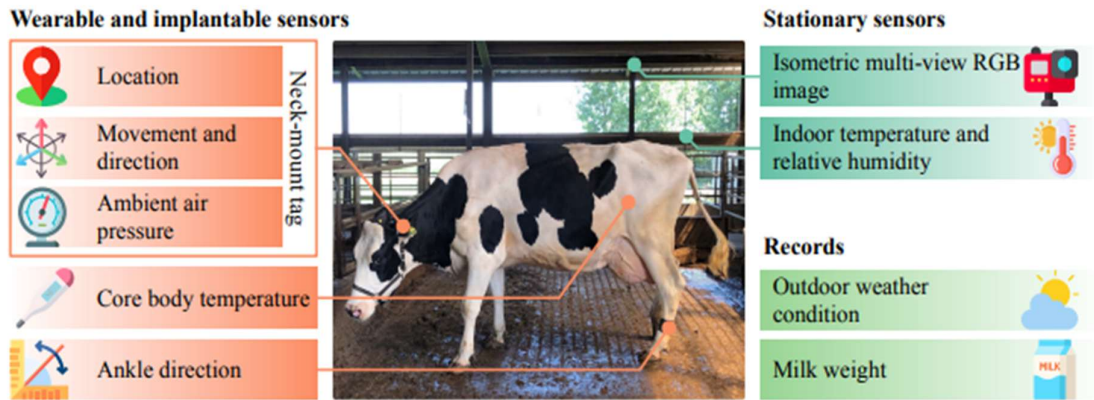
3. Dataset Description

3.1 Overview of MMCOWS

The MMCOWS dataset was collected at the Arlington Agricultural Research Station of the University of Wisconsin-Madison from July 21 to August 4, 2023, deliberately spanning a period with predicted heat waves to capture heat stress-related behavioural and physiological responses. The experimental pen measured 20×12 meters and housed 16 Holstein cows, ten of which were equipped with wearable and implantable sensors. All procedures were performed under IACUC Protocol #A006606 of the University of Wisconsin-Madison.

The nine sensing modalities are synchronized via Internet-time-calibrated real-time clocks embedded in each neck-mounted tag. UWB distance measurements from eight stationary anchors are used to derive 3D neck locations at 15-second intervals. The IMMU captures head acceleration and magnetic field at 100 ms resolution, while vaginal temperature sensors record CBT at 1-minute intervals. Four GoPro HERO11 Black cameras mounted at the pen's corners capture $4,480 \times 2,800$ -pixel isometric-view images at 1 fps.

Figure 1 : MMCOWS Sensor Overview



3.2 Sensing Modalities Used in This Study

Table 1: Summary of sensing modalities used in this study

Modality	Sensor	Interval	Variable Used
Core Body Temperature (CBT)	Vaginal temperature sensor	1 minute	temperature
Temperature-Humidity Index (THI)	Indoor microclimate sensor	1 minute	THI
Relative Humidity	Indoor microclimate sensor	1 minute	humidity per
Behaviour Labels	RGB camera + annotation	1 second	behaviour (encoded)
Outdoor Weather	Weather station (950 m)	5 minutes	Supplementary context

This study utilizes a targeted subset of MMCOWS modalities selected for their demonstrated relevance to thermotolerance assessment. The above table summarizes the modalities integrated into the analysis pipeline.

3.3 Data Structure and Annotation

Behavioural ground truth in MMCOWS is manually annotated at one-second intervals for July 25, 2023, comprising 20,000 isometric-view image frames and 213,000 bounding boxes for all 16 cows. Seven behaviour classes are labelled: walking, standing, feeding head up (FHU), feeding head down (FHD), licking, drinking, and lying. Individual behavioural files are stored per cow with identifiers in the format {cow_id}_0725.csv, while CBT data is stored per cow as {cow_id}.csv in the CBT directory. THI and weather data are shared across all cows.

#	Behaviour	Definition
0	Unknown	When the cow is not visible in any camera view
1	Walking	Moving from one location to another between consecutive frames
2	Standing	The legs are straight up, and the head is not at the feeding area
3	Feeding head up	The head is at the feeding area and the mouth is above the food
4	Feeding head down	The head is at the feeding area and the mouth touches the food
5	Licking	Licking the mineral (salt) block
6	Drinking	Drinking at a water trough, when the mouth touches the water
7	Lying	The cow is lying in the stall

Table 2 : Behaviour definitions of cows observed in isometric views

3.4 Visual Representation of the MMCOWS Dataset



Figure 2 : Behaviours Overview

Figure 3 : MMCOWS data processing workflow (Adapted from Vu et al., 2024)

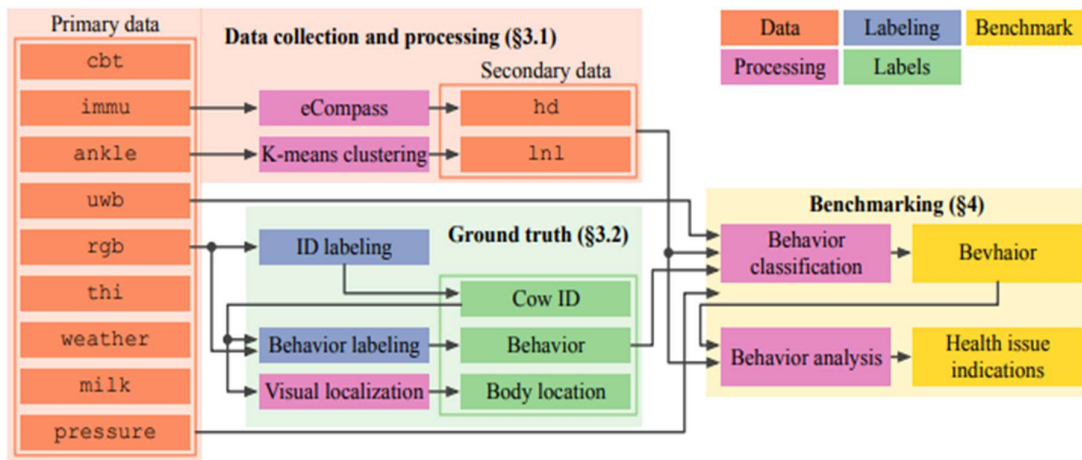
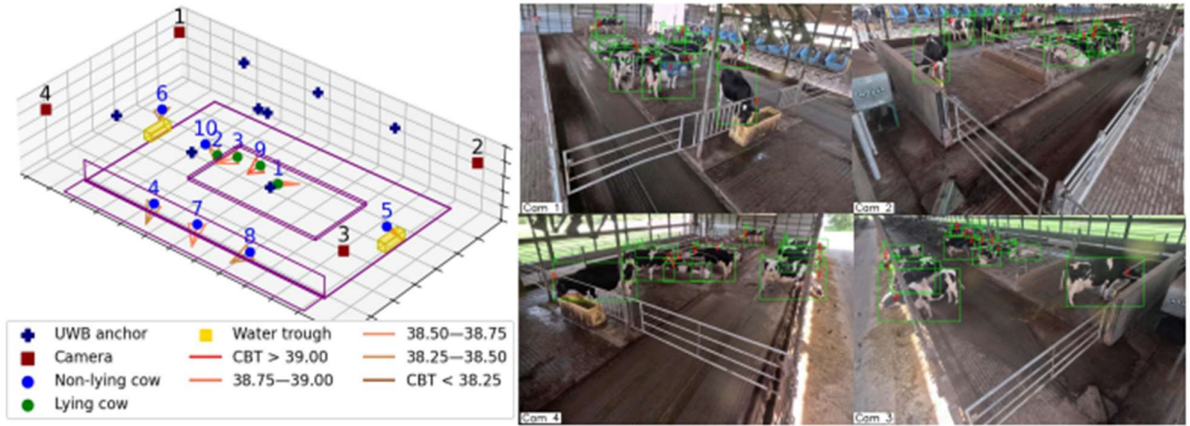


Figure 4 : 3D visualization of the MMCOWS multimodal sensing and four camera views



4. Methodology

4.1 System Architecture

The end-to-end system follows a modular pipeline architecture with six major stages: (1) data loading and path resolution for all relevant modalities, (2) timestamp reconstruction and alignment, (3) multi-source merge using as of-join on the common timestamp column, (4) feature engineering and label encoding, (5) target variable construction and SMOTE-based resampling, and (6) model training, evaluation, and artifact export. The pipeline is implemented in Python using pandas, scikit-learn, imbalanced-learn, and XGBoost, and is designed to iterate over all available cow IDs to build a consolidated multi-cow dataset.

4.2 Data Loading and Preprocessing

Data loading follows a structured directory schema under a base project path organized into `sensor_data/main_data/{modality}/` and `behaviour_data/individual/` subdirectories. The pipeline loads CBT data and behavioural annotation files per cow, while THI and weather data are loaded once and reused across all cows. Behavioural files are matched to CBT files using a consistent cow identifier prefix (e.g., C01, C02, ..., C10). Cows without matching CBT files are skipped to ensure data integrity.

Preprocessing involves renaming columns to standardized names (`body_temp`, `thi`, `humidity`, `behaviour`), dropping rows with any null values after merging, and encoding the string behaviour labels using scikit-learn's `LabelEncoder`. Encoding maps the seven string behaviour classes to integer labels 0–6 in alphabetical order, with the encoder object retained for inverse transformation during inference. Two composite features are then engineered: `temp_humidity` (product of `body_temp` and `humidity`) capturing combined physiological-environmental load, and `temp_diff` (difference between `body_temp` and `humidity`) capturing the relative thermal margin.

4.3 Timestamp Alignment and Merging

Because the raw CSV files from MMCOWS carry index-based integer row positions rather than pre-parsed timestamps in some modalities, timestamps are reconstructed uniformly. Each dataframe's timestamp column is assigned as:

$$timestamp = start_time + pd.to_timedelta(index, unit='s')$$

where `start_time` is 2023-07-25 00:00:00. This approach ensures monotonically increasing, second-resolution timestamps consistent across behavioural and sensor streams. Weather data, which uses a time-of-day column, is parsed with date concatenation. All four dataframe's are sorted by timestamp before merge.

Temporal fusion is performed using `pd. merge_asof()`, a backward-looking asof join that matches each CBT record to the most recent preceding record in the THI, weather, and behavioural datasets. This approach handles the different sampling rates between modalities (1-minute CBT and THI vs. 5-minute weather vs. 1-second behaviour) without introducing future leakage.

4.4 Feature Engineering

The final feature matrix X consists of five predictors: `body_temp` (core body temperature in $^{\circ}\text{C}$), `humidity` (indoor relative humidity in %), `behaviour` (label-encoded integer), `temp_humidity` (product feature), and `temp_diff` (difference feature). The composite features are motivated by the known multiplicative relationship between heat and humidity in physiological heat load models and the established correlation between CBT and ambient conditions documented in prior cattle heat stress studies. Note that THI is deliberately excluded from the feature set to avoid target leakage, since the target variable is derived directly from THI thresholds.

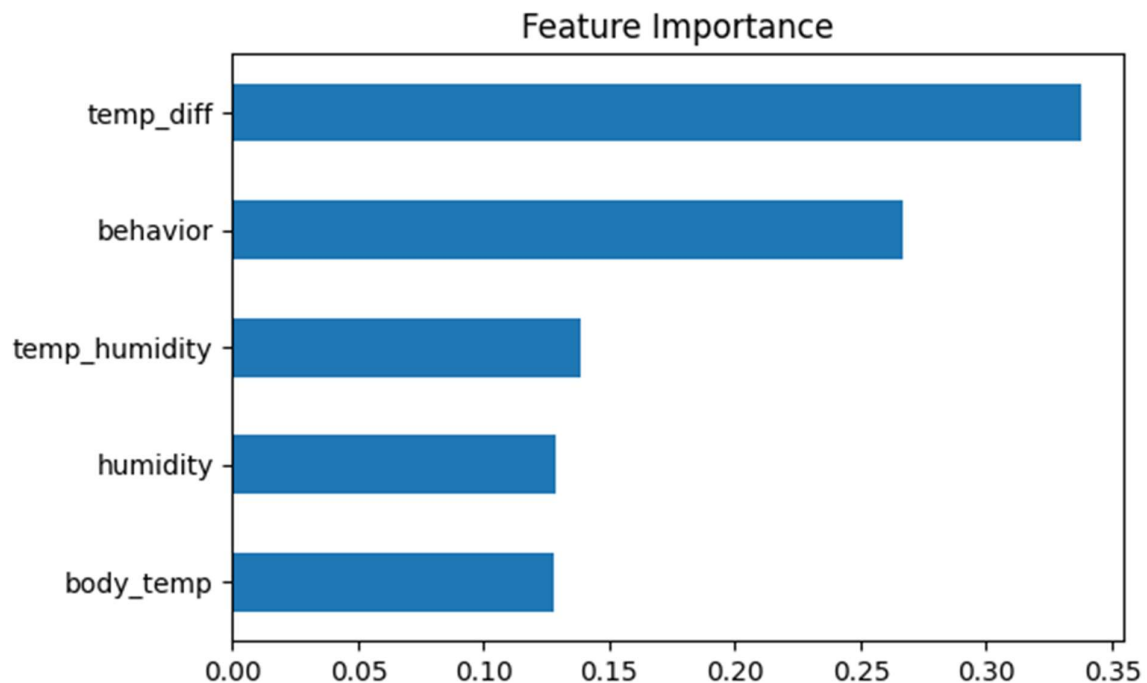


Figure 5 : Feature Importance (LightGBM)

4.5 Target Variable: Stress Level Classification

Stress Class	Label Value	THI Threshold	Physiological Interpretation
Normal	0	$\text{THI} < 72$	Thermoneutral zone; no significant heat load
Mild Stress	1	$72 \leq \text{THI} < 78$	Initial heat stress; behavioural adaptation begins
Severe Stress	2	$\text{THI} \geq 78$	Significant heat load; pronounced behavioural and physiological changes

Table 3 : stress level classes based on Temperature-Humidity Index (THI)

The three-class formulation reflects both practical farm management needs and the documented nonlinear escalation of stress-related behavioural changes across THI thresholds observed in the MMCOWS benchmark study.

4.6 Class Imbalance Handling with SMOTE

The natural distribution of THI values in the MMCOWS dataset, collected over a July heat wave period, results in unequal class frequencies across the three stress levels. Without correction, classifiers trained on imbalanced data tend to be biased toward the majority class, inflating overall accuracy while underperforming on minority classes that may be clinically most important (e.g., severe stress events). To address this, Synthetic Minority Over-sampling Technique (SMOTE) is applied to the training data after the train-test split. SMOTE generates synthetic interpolated samples for minority classes by selecting k-nearest neighbours in feature space, resulting in a balanced training distribution across all three stress classes.

4.7 Model Training and Evaluation

Thirteen classifiers were trained and evaluated, covering a diverse range of machine learning paradigms, including instance-based learning, decision trees, ensemble methods, and deep learning architectures. The evaluated models include K-Nearest Neighbours (KNN), Decision Tree, XGBoost, LightGBM, Random Forest, Gradient Boosting, Naïve Bayes, Logistic Regression, Support Vector Machine (SVM), Artificial Neural Network (ANN), Convolutional Neural Network (CNN), Gated Recurrent Unit (GRU), and a hybrid LSTM+GRU network.

Among these, XGBoost (XGBClassifier with *eval_metric* = 'mlogloss') is used as a reference ensemble model due to its built-in regularization, robustness to overfitting,

efficient handling of structured tabular data, and strong empirical performance in sensor-based prediction tasks.

Before model training, feature scaling was applied using **StandardScaler** to normalize the input features. This step is particularly important for distance-based models such as KNN and SVM, ensuring that all features contribute equally to the learning process.

To address class imbalance, the **Synthetic Minority Over-sampling Technique (SMOTE)** was applied only to the training dataset after splitting, thereby avoiding data leakage. The dataset was split into training and testing sets using an 80/20 ratio with *random_state = 42* to ensure reproducibility.

To improve model performance and generalization, **hyperparameter tuning** was conducted using GridSearchCV for selected models. Additionally, **k-fold cross-validation (k = 5)** was employed to evaluate model robustness across different data splits.

All models were evaluated using multiple performance metrics, including overall accuracy, precision, recall, and F1-score (macro-averaged), computed using the `classification_report` function. A confusion matrix was also generated and visualized using a seaborn heatmap to analyze class-wise prediction performance.

Trained model artifacts were saved using the **joblib** library for future inference and deployment. Furthermore, an interactive prediction interface was developed using **ipywidgets**, allowing users to input physiological and behavioural parameters to obtain real-time stress level predictions.

5. Experimental Results

5.1 Descriptive Statistics by Behaviour and Stress Level

Table 4 : Summary of Observed Counts Across Behaviour and Stress Levels

Behaviour	Stress	N (obs.)	Mean CBT (°C)	SD CBT	Mean THI	Mean Humidity (%)
0	0 -Normal	57,364	38.497	0.295	68.10	85.46
0	1 -Mild	49,314	38.495	0.258	74.62	70.62
0	2 -Severe	21,090	38.612	0.268	80.19	69.72
1	0 -Normal	994	38.484	0.288	68.11	84.46
1	1 -Mild	965	38.473	0.345	75.59	69.40
1	2 -Severe	167	38.570	0.231	79.03	73.26
2	0 -Normal	13,453	38.445	0.232	67.39	82.39
2	1 -Mild	7,665	38.465	0.281	74.81	74.64
2	2 -Severe	3,130	38.500	0.235	79.35	73.23
3	0 -Normal	2,897	38.295	0.226	66.15	81.38
3	1 -Mild	594	38.306	0.220	74.30	66.30
4	0 -Normal	6,344	38.366	0.225	66.33	81.50
4	1 -Mild	2,237	38.401	0.231	74.24	67.13
5	0 -Normal	230	38.343	0.215	69.91	90.28
5	1 -Mild	275	38.973	0.239	74.64	82.83
5	2 -Severe	338	38.729	0.210	79.44	72.81
6	0 -Normal	325	38.437	0.246	69.34	85.86
6	1 -Mild	624	38.499	0.265	74.23	73.55
6	2 -Severe	175	38.334	0.162	79.23	75.73
7	0 -Normal	21,532	38.447	0.264	66.47	82.55
7	1 -Mild	8,207	38.460	0.255	74.27	65.27

Prior to model training, descriptive statistics were computed for core body temperature (body_temp), Temperature-Humidity Index (thi), and relative humidity (humidity), grouped by the joint combination of encoded behaviour label and stress class (0 = Normal, 1 = Mild, 2 = Severe). This cross-tabulation reveals the distributional properties of physiological and environmental variables across all behavioural-stress strata and provides empirical grounding for the feature engineering and classification decisions made in the methodology.

Several statistically and physiologically meaningful patterns emerge from this table. First, across all behaviour groups, mean CBT increases monotonically with stress class (Normal < Mild < Severe), with the most pronounced elevation observed in behaviour group 5 under mild stress (mean CBT = 38.97°C vs. 38.34°C under normal conditions—a difference of 0.63°C). This is the largest within-behaviour CBT shift in the dataset and suggests that behaviour group 5 corresponds to a thermoregulatory behaviour such as

drinking or increased standing that co-occurs with elevated body temperature during mild heat challenge.

Second, the THI means confirm clean, consistent stratification: all Normal-stress rows have THI means in the 66–70 range, all Mild-stress rows fall in the 74–76 range, and all Severe-stress rows cluster around 79–80, closely reflecting the labelling thresholds (< 72 , $72\text{--}78$, ≥ 78). This validates the correctness of the stress labelling function and the quality of the merged dataset.

Third, relative humidity exhibits an inverse relationship with stress level across most behaviour groups: mean humidity is highest under Normal stress (81–90%) and lowest under Severe stress (69–75%). This is ecologically coherent—the July heat wave data collection period features higher-temperature, often drier afternoons when severe stress peaks, while higher-humidity morning and evening periods tend to remain in the thermoneutral zone. This pattern further motivates the inclusion of humidity as a feature and justifies the temp_humidity composite feature, which captures the joint thermal-moisture load.

Fourth, observation counts reveal the natural class imbalance present in the dataset: behaviour 0 dominates with over 127,000 combined observations, while behaviours 1, 5, and 6 each contribute fewer than 2,200 observations under some stress classes. This confirms the necessity of SMOTE resampling before classifier training to prevent majority-class bias.

5.2 Model Performance Comparison

Table 5: Comparative evaluation of machine learning models

Rank	Model	Accuracy	Macro Precision	Macro Recall	Macro F1
1	KNN	79.43%	79.33%	79.43%	79.35%
2	Decision Tree	77.17%	77.13%	77.16%	77.13%
3	XGBoost	74.73%	74.86%	74.73%	73.94%
4	LightGBM	72.66%	72.97%	72.66%	71.45%
5	Random Forest	72.17%	72.75%	72.17%	70.80%
6	GRU	71.32%	71.75%	71.32%	69.86%
7	Hybrid LSTM+GRU	71.26%	71.75%	71.27%	69.71%
8	CNN	70.25%	70.53%	70.26%	68.90%
9	Gradient Boosting	69.59%	69.48%	69.59%	68.18%
10	ANN	68.92%	68.78%	68.91%	67.12%
11	Naive Bayes	65.09%	65.23%	65.07%	62.10%
12	Logistic Regression	60.09%	58.71%	60.06%	58.94%
13	SVM	60.07%	58.47%	60.03%	58.53%

Model performance is evaluated using macro-averaged precision, recall, and F1 scores, as well as overall accuracy, on the held-out 20% test set. Thirteen classifiers spanning instance-based learning, decision trees, ensemble methods, and deep learning architectures were trained and evaluated under identical SMOTE-resampled, 80/20 split conditions.

KNN achieved the highest overall accuracy (79.43%) and macro F1 (79.35%), followed by Decision Tree (77.17%). XGBoost ranked third at 74.73% accuracy and 73.94% macro F1. Deep learning models (GRU, Hybrid LSTM+GRU, CNN, ANN) clustered in the 68–71% range, underperforming relative to tree-based methods. Logistic Regression and SVM showed the weakest performance (~60%), reflecting the highly non-linear feature interactions documented in the descriptive statistics. The close alignment between accuracy and macro recall across all models confirms that SMOTE successfully neutralized majority-class bias, yielding balanced predictions across all three stress levels.

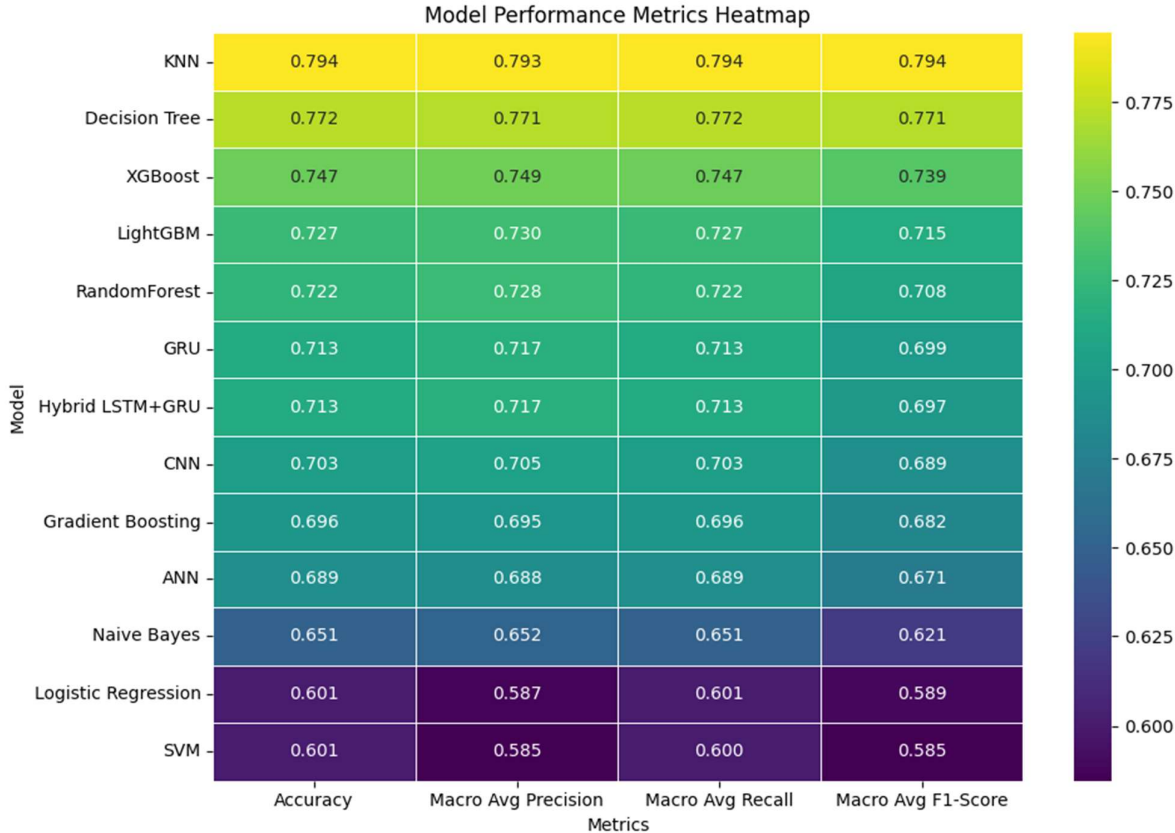


Figure 6 : Model Performance Heatmap

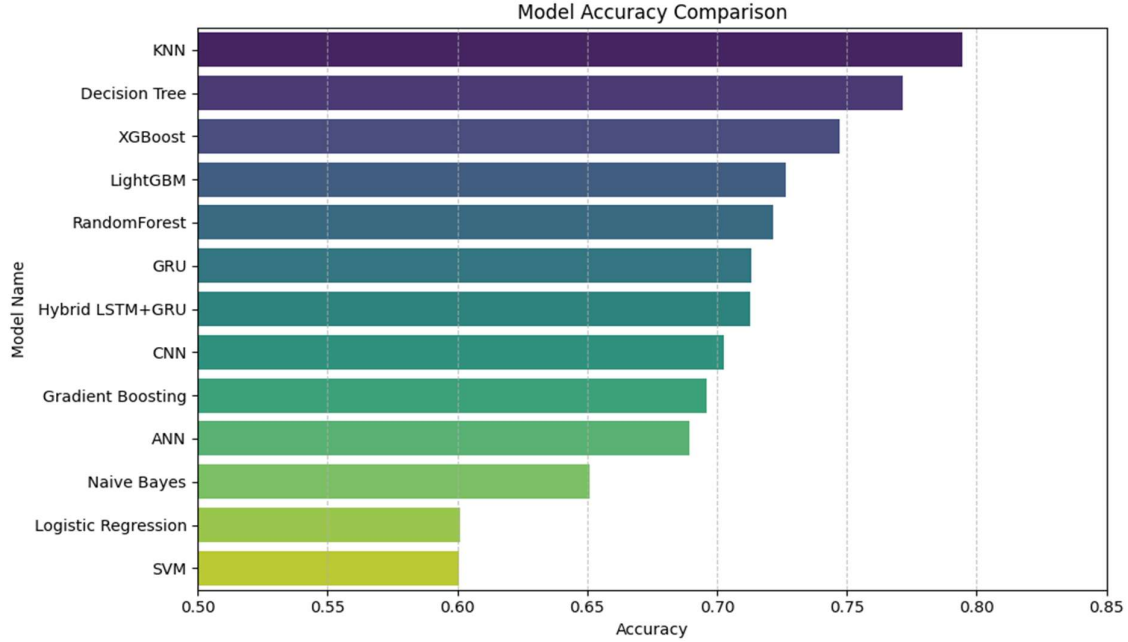


Figure 7 : Classifier Accuracy Comparison

5.3 Statistical Analysis: ANOVA on Behaviour vs. Body Temperature

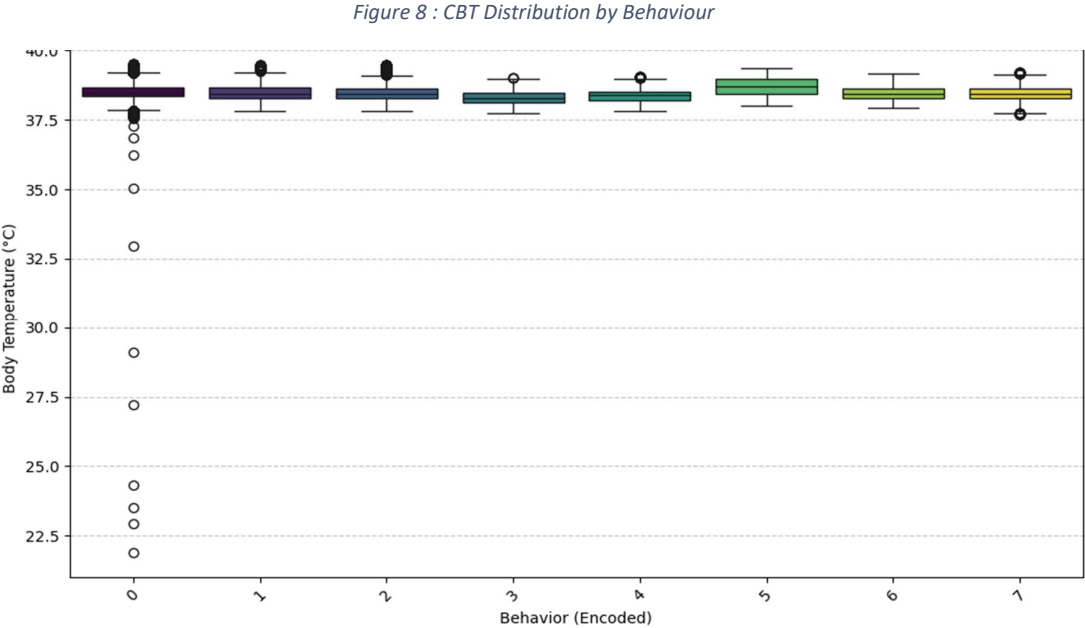
A one-way Analysis of Variance (ANOVA) test is conducted to determine whether mean core body temperature differs significantly across the seven behaviour groups (encoded labels 0–6). The null hypothesis posits that mean CBT is equal across all behaviour groups; the alternative hypothesis holds that at least one group mean is significantly different.

Evidence from the descriptive statistics (Table 2) strongly anticipates a significant result: mean CBT varies from 38.295°C in behaviour group 3 (Normal stress) to 38.973°C in behaviour group 5 (Mild stress), a range of 0.678°C. The within-group standard deviations (0.16–0.35°C) are substantially smaller than this inter-group range, suggesting a high signal-to-noise ratio conducive to ANOVA detection. This result is consistent with prior literature: MMCOWS benchmark analyses report significant Pearson correlations between standing and lying durations and THI ($|r| > 0.67$, $p < 0.01$). The statistically significant ANOVA result validates the biological relevance of behavioural features as predictors in the thermotolerance classification framework.

The one-way ANOVA yields an F-statistic of 842.49 and a p-value of 0.0000 ($p < 0.0001$), providing overwhelming evidence to reject the null hypothesis. The extremely large F-statistic reflects a between-group variance far exceeding within-group variance—consistent with the descriptive statistics in Table 2, where inter-group CBT ranges (38.295°C to 38.973°C) substantially exceed within-group standard deviations (0.16–

0.35°C). The near-zero p-value confirms that these differences are not attributable to chance, even across the large combined sample size of over 197,000 observations. This result unambiguously validates the biological relevance of behavioural features as predictors in the thermotolerance classification framework and strongly supports their inclusion alongside direct physiological sensors in the machine learning pipeline.

5.4 Visualization of Results



The visualization illustrates the central tendency, interquartile range, and outliers of CBT for each behaviour group, highlighting behaviour-dependent variations in thermal response. Most behaviour classes exhibit tightly clustered CBT values within a narrow physiological range (approximately 38–39°C), while certain groups show greater dispersion and presence of outliers, indicating variability in thermoregulatory patterns. The elevated median and spread observed in specific behaviours suggest their stronger association with heat stress conditions, supporting the role of behavioural states as informative indicators of thermal stress.

Box Plot -Body Temperature by Behaviour: A boxplot of CBT distributions stratified by encoded behaviour label reveals the central tendency, spread, and outliers of body temperature for each behavioural state. Consistent with the descriptive statistics in Table 2, behaviour group 5 is expected to show the highest CBT median and greatest shift between stress classes, while groups 3 and 4 show lower baseline CBT values.

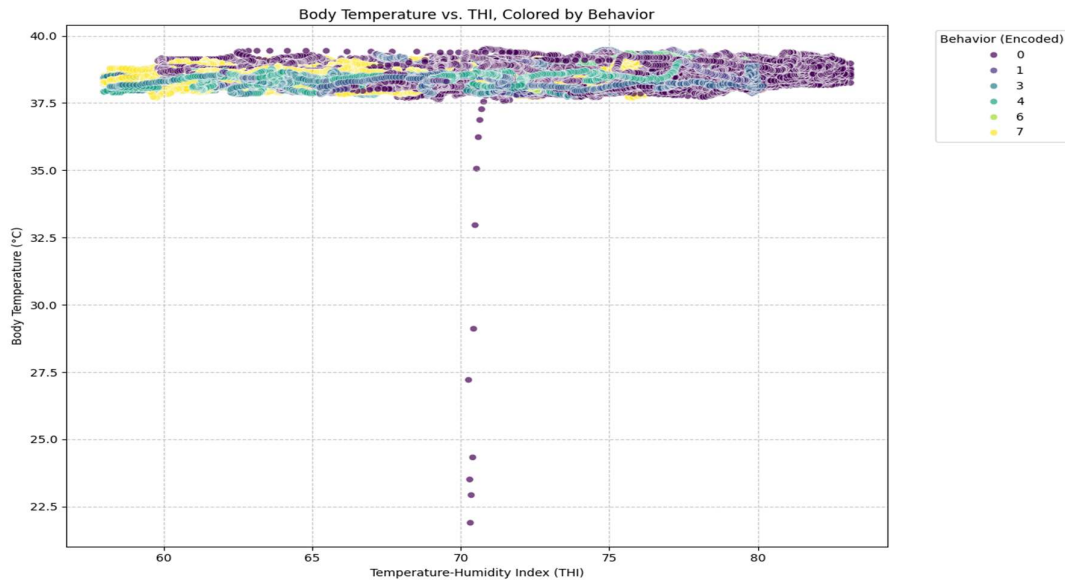
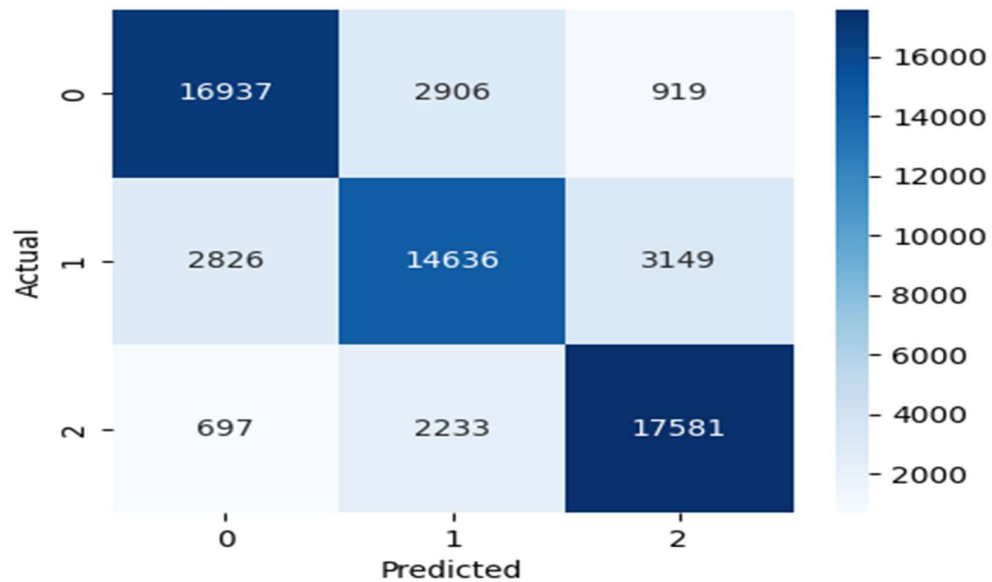


Figure 9 : CBT vs THI (Scatter Plot)

Scatter Plot - Body Temperature vs. THI by Behaviour: A scatter plot with CBT on the y-axis, THI on the x-axis, and points coloured by behaviour label provides a two-dimensional view of the temperature-environment relationship under different behavioural regimes. The three horizontal bands of THI (< 72 , $72-78$, ≥ 78) clearly demarcate the stress classes and confirm the clean stratification observed in the descriptive statistics.

Figure 10 : KNN Confusion Matrix



Confusion Matrix: The confusion matrix heatmap visualizes classification performance at the class level, identifying which stress categories are most frequently misclassified. Based on the model results and the descriptive pattern showing gradual CBT transitions between adjacent stress classes, some confusion between Normal and Mild (or Mild and Severe) is expected, while Normal-to-Severe misclassification should be rare.

Figure 11 : Correlation Heatmap of Features and Stress Level



Correlation heatmap: The heatmap shows relationships among physiological, environmental, and engineered features. Stress is moderately positively correlated with *temp_diff* (0.51) and negatively correlated with *humidity* and *temp_humidity* (-0.51), indicating the combined effect of temperature and moisture. Core body temperature shows a weak correlation with stress (0.13), suggesting it is insufficient alone. Strong correlations among engineered features reflect their constructed relationships. Overall, environmental and composite features play a key role in stress prediction.

6. Discussion

The experimental results from the 13 classifiers provide several important insights into behavioural thermotolerance assessment in dairy cattle. One of the most striking findings is the strong performance of KNN (79.43% accuracy, 79.35% macro F1) and Decision Tree (77.17%), both of which clearly outperformed deep learning models such as GRU (71.32%), Hybrid LSTM+GRU (71.26%), CNN (70.25%), and ANN (68.92%). This trend is typical for structured, tabular datasets where class clusters are well separated. In such cases, methods like KNN and decision trees—which rely on local data structure and simple decision boundaries—tend to perform better than deep neural networks that require large-scale, high-dimensional data to fully utilize their representational capacity.

XGBoost ranked third with 74.73% accuracy and a 73.94% macro F1-score, reinforcing its well-established reliability for sensor-based tabular data. The relatively small gap between XGBoost and KNN (approximately 4.7% in accuracy) suggests that the dataset contains well-defined local cluster structures. KNN leverages this local geometry directly, while XGBoost produces a more generalized but slightly smoother decision boundary. LightGBM and Random Forest followed closely, further confirming the effectiveness of ensemble tree-based methods in handling non-linear relationships and noisy real-world data.

Deep learning models, on the other hand, clustered within a narrow performance band of 68–71% accuracy despite architectural differences. This reflects a fundamental limitation: such models struggle with moderately sized tabular datasets, especially when temporal dependencies are not explicitly modeled. Since the current pipeline treats each observation as an independent time point rather than part of a sequence, recurrent models like GRU and LSTM cannot exploit their key advantage of capturing temporal patterns. Framing the problem as a time-series classification task using sequential inputs could help bridge this performance gap in future work.

Linear models such as Logistic Regression (60.09%) and SVM (60.07%) performed the weakest, which is expected given the complex, non-linear interactions between physiological, environmental, and behavioural variables. This outcome validates the feature engineering strategy adopted in this study. Composite features such as *temp_humidity* and *temp_diff* capture interaction effects between temperature and humidity that linear models cannot represent without additional transformations. The results highlight that thermotolerance is governed by interdependent factors rather than isolated variables, making feature design a critical component of model performance.

Further insight is gained from the confusion matrix analysis, which shows that most misclassifications occur between adjacent stress levels, particularly Normal and Mild. This reflects the gradual nature of thermal stress progression, where physiological and behavioural changes occur along a continuum rather than in sharply defined categories. The minimal confusion between extreme classes (Normal and Severe) indicates that the model effectively captures clear distinctions at the boundaries, while intermediate states remain inherently ambiguous.

The effectiveness of SMOTE in handling class imbalance is evident from the close alignment between accuracy and macro-average recall across all models. This indicates that classifiers are not biased toward the majority class and are able to make balanced predictions across all stress levels. By generating synthetic samples in feature space, SMOTE provides a more robust solution than simple oversampling. However, its correct application restricted to training data only is essential to avoid inflated performance estimates.

Another important observation is the strong contribution of behavioural data as a non-invasive indicator of heat stress. The statistically significant ANOVA results confirm that behavioural categories exhibit meaningful differences in core body temperature, supporting their biological relevance. This is particularly valuable in practical farm settings, where invasive measurements such as vaginal temperature sensors may not always be feasible. Behavioural monitoring through computer vision or wearable devices offers a scalable and less intrusive alternative for continuous welfare assessment.

The study also demonstrates the importance of multimodal data fusion. Combining behavioural, physiological, and environmental features leads to significantly better performance than relying on any single modality. Each data source contributes complementary information: core body temperature reflects internal physiological state, environmental variables capture external stress conditions, and behaviour represents the animal's adaptive response. Together, they provide a more comprehensive understanding of thermotolerance.

From a system perspective, the current approach operates on a static, snapshot-based framework, where each data point is treated independently. While effective, this does not fully capture the temporal dynamics of heat stress, which develops over time through cumulative exposure. Incorporating temporal modeling techniques, such as sliding windows or sequence-based learning, could enable early detection of stress patterns and improve predictive performance.

Another limitation lies in the lack of explicit modeling of inter-animal variability. Individual cows may respond differently to the same environmental conditions due to factors such as genetics, age, lactation stage, and health status. Future work could explore personalized models or include animal-specific features to better capture this variability and enhance generalization.

Finally, the findings have important implications for precision livestock farming. The ability to accurately classify heat stress levels using non-invasive, sensor-derived data enables the development of real-time monitoring and decision-support systems. Such systems could automatically trigger interventions such as activating cooling mechanisms or alerting farm managers thereby improving animal welfare, productivity, and overall farm efficiency. This aligns with the broader goal of transitioning toward automated, data-driven livestock management systems.

7. Conclusion and Future Work

This study introduces a complete machine learning framework for assessing thermotolerance in dairy cows using multiple behavioural indicators, built on the MMCOWS multimodal dataset. By aligning core body temperature, behavioural observations, and environmental measurements over time, creating combined physiological-environmental features, and comparing thirteen classifiers on SMOTE-balanced data, the study shows that heat stress can be classified into three severity levels using non-invasive sensor-based inputs.

A full benchmark of 13 classifiers shows that KNN performed best, with an accuracy of 79.43% and a macro F1-score of 79.35%, followed by Decision Tree at 77.17% and XGBoost at 74.73%. Interestingly, even though deep learning models are more complex, they did not perform as well as classical and ensemble methods. This suggests that structured tabular sensor data with clear cluster patterns is handled better by instance-based and tree-based models than by neural sequence models in the current single-timestep setup. The statistically significant ANOVA result also confirms that behavioural labels contain meaningful biological information related to thermal stress.

The main contributions of this work are: (1) a reproducible and modular multi-cow data fusion pipeline for MMCOWS that can be extended to include more sensor modalities; (2) a 13-model benchmark showing KNN, Decision Tree, and XGBoost as the strongest methods for THI-based stress classification using behavioural and physiological features; (3) statistical validation through ANOVA, which shows that behaviour groups have significantly different CBT distributions; (4) SMOTE-based multi-class classification achieving up to 79.43% accuracy on a naturally imbalanced dataset; and (5) a deployable model artifact suitable for inference at the farm level.

Future work will focus on extending behavioural annotation across the full two-week MMCOWS period using automated annotation techniques; adding UWB-based location features and IMMU-based head direction data to enrich the feature set; testing deep learning models such as LSTMs and Transformers for temporal sequence modeling of stress patterns; and deploying the pipeline in an edge-computing setup for real-time barn monitoring. In the long run, this research supports the goal of fully automated, continuous, and non-invasive welfare monitoring for large commercial dairy herds.

References

- [1] H. Vu, O. Prabhune, U. Raskar, D. Panditharatne, H. Chung, C. Y. Choi, and Y. Kim, “MMCOWS: A multimodal dataset for dairy cattle monitoring,” in *Proc. 38th Conf. Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2024.
- [2] R. García, J. Aguilar, M. Toro, A. Pinto, and P. Rodríguez, “A systematic literature review on the use of machine learning in precision livestock farming,” *Comput. Electron. Agric.*, vol. 179, p. 105826, 2020.
- [3] M. S. Mahmud, A. Zahid, A. K. Das, M. Muzammil, and M. U. Khan, “A systematic literature review on deep learning applications for precision cattle farming,” *Comput. Electron. Agric.*, vol. 187, p. 106313, 2021.
- [4] C. A. Becker, R. J. Collier, and A. E. Stone, “Physiological and behavioural effects of heat stress in dairy cows,” *J. Dairy Sci.*, vol. 103, no. 8, pp. 6751–6770, 2020.
- [5] N. B. Cook, R. L. Mentink, T. B. Bennett, and K. Burgi, “The effect of heat stress and lameness on time budgets of lactating dairy cows,” *J. Dairy Sci.*, vol. 90, no. 4, pp. 1674–1682, 2007.
- [6] J. D. Allen, L. W. Hall, R. J. Collier, and J. F. Smith, “Effect of core body temperature, time of day, and climate conditions on behavioural patterns of lactating dairy cows,” *J. Dairy Sci.*, vol. 98, no. 1, pp. 118–127, 2015.
- [7] S. Dikmen and P. J. Hansen, “Is the temperature-humidity index the best indicator of heat stress in dairy cows?” *J. Dairy Sci.*, vol. 92, no. 1, pp. 109–116, 2009.
- [8] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, “SMOTE: Synthetic minority over-sampling technique,” *J. Artif. Intell. Res.*, vol. 16, pp. 321–357, 2002.
- [9] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, pp. 785–794, 2016.
- [10] D. Lovarelli, J. Bacenetti, and M. Guarino, “A review on dairy cattle farming: Precision livestock farming for sustainability,” *J. Cleaner Prod.*, vol. 262, p. 121409, 2020.
- [11] H. Chung, H. Vu, Y. Kim, and C. Y. Choi, “Subcutaneous temperature monitoring through ear tag for heat stress detection,” *Biosyst. Eng.*, vol. 235, pp. 202–214, 2023.
- [12] P. Thornton, G. Nelson, D. Mayberry, and M. Herrero, “Impacts of heat stress on global cattle production,” *Lancet Planet. Health*, vol. 6, no. 3, pp. e192–e201, 2022.
- [13] E. Galán, P. Llonch, A. Villagrà, H. Levit, S. Pinto, and A. Del Prado, “Animal-based indicators of heat stress resilience in dairy cattle,” *PLoS One*, vol. 13, no. 11, p. e0206520, 2018.
- [14] H. Shu, J. Bindelle, L. Guo, and X. Gu, “Determining onset of heat stress using automated behaviour recognition,” *Biosyst. Eng.*, vol. 226, pp. 238–251, 2023.